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Passive cavitation mapping for localization and tracking of bubble dynamics

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Abstract: Current acoustic techniques for studying cavitation dynamics are only readily applicable to single-bubble activity, while optical methods can only be used in transparent media. However, multi-bubble cavitation often occurs in opaque media such as biological tissue. Here, the signals received passively by each of the 64 channels of a diagnostic ultrasound array are used to localize and separate emissions from several bubble clusters cavitating in agar gel, thereby providing a method of observing cavitation dynamics. The method has a high spatiotemporal resolution and is applicable to cavitation in opaque media.

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1. Introduction

Cavitation—the linear and nonlinear oscillations of bubbles in response to external stimuli such as an acoustic field—is receiving increased attention as a phenomenon involved in many ultrasound therapies.¹ If treatment protocols are to ensure efficiency and selectivity, it is important to establish which types of cavitation are associated with different bioeffects and how the spatial distribution of different types of cavitation can be expected to vary with time.

High-speed photography (HSP) and the use of passive cavitation detection (PCD) have brought major advances in our understanding of cavitation dynamics and associated bioeffects. HSP can capture bubbles at megahertz frame rates,² allowing the observation of biologically relevant effects such as streaming² and jetting near a solid boundary.^{3,4} However, in addition to its high cost, HSP is unable to image inside opaque media such as most human tissue, which calls for alternative methods of cavitation detection. PCD detects bubble activity by sensing the acoustic emissions of cavitating bubbles.⁵ Because the far-field radiated pressure of a bubble is proportional to the second temporal derivative of its volume, spectral peaks in volumetric oscillations register as spectral peaks in pressure emissions, yielding information about the periodicity and stability of bubble motion.⁶ However, despite its ability to provide spatially-dependent sensitivity on receive, conventional PCD using a single-element transducer cannot separate the mixture of emission traces it receives based on source location. As a result, the received signal could lead to erroneous conclusions about the types, locations, and mutual interactions of cavitating bubbles. Spatial registration between cavitation and bioeffects is therefore not presently possible.

Recently, diagnostic ultrasound systems with access to post-beamformed RF data have been used to provide spatial information about cavitation activity. **By turning off pulse transmission, the dynamic receive beamforming algorithm of the ultrasound scanners has been used to resolve cavitation along the transverse direction of the linear array,^{5,7} and, to a limited extent, axially.⁸** However, the reliance on sequentially generated RF A-lines precludes these methods from simultaneously detecting emissions from spatially distinct cavitation sources. Moreover, the fixed size of the subapertures used to form the A-lines greatly degrades the axial resolution of the images. Here, we show how a custom passive beamforming algorithm applied to pre-beamformed receiver data (“channel data”) is able to overcome these difficulties.

A system to monitor the spatial distribution of cavitation during high intensity focused ultrasound (HIFU) treatment in a clinically relevant setup, whereby the array of detectors is placed co-axially with the HIFU transducer, has already been presented elsewhere.^{9,10} Building on this work, we focus in the present article on the unique ability of the passive beamforming algorithm to separate cavitation emissions from a population of bubbles and thereby observe the interactions between individual bubbles or bubble clusters. The array is placed here at 90° to the HIFU beam, in order to maximize the resolution for this particular array transducer, but it should be noted that the method is equally applicable in a coaxial setup.

2. Theory

Bubbles undergoing radial oscillations can be characterized by their source strength, commonly defined as the rate of change of mass outflow generated by the bubble (expressed in subsequent figures in $\mu\text{g } \mu\text{s}^{-2}$). For a bubble centered at $\mathbf{r}$ and with a source strength of $q(\mathbf{r}, t)$, the resulting pressure at a location $\mathbf{r}'$ and time t is given by¹¹

$$p(\mathbf{r}', t) = \frac{q(t - |\mathbf{r}' - \mathbf{r}|/c)}{4\pi|\mathbf{r}' - \mathbf{r}|}, \quad (1)$$

where c is the speed of sound of the medium. Using an array of calibrated pressure sensors at locations $\mathbf{r}_1, \dots, \mathbf{r}_N$ with equal noise characteristics, Eq. (1) allows an estimate $\tilde{q}(\mathbf{r}, t)$ of the source strength, and by extension, the acoustic power $\tilde{P}(\mathbf{r}, t)$ radiated by the bubble:

$$\tilde{q}(\mathbf{r}, t) = \frac{1}{N} \sum_{i=1}^N 4\pi|\mathbf{r}_i - \mathbf{r}| \tilde{p}(\mathbf{r}_i, t + |\mathbf{r}_i - \mathbf{r}|/c), \quad (2)$$

$$\tilde{P}(\mathbf{r}, t) = \frac{\langle \tilde{q}^2(\mathbf{r}, t) \rangle}{4\pi\rho_0 c}, \quad (3)$$

where $\tilde{p}(\mathbf{r}_i, t)$ is the pressure at $\mathbf{r}_i$ estimated from the corresponding pressure sensor, $\langle \cdot \rangle$ is the mean operator, and ρ_0 is the (undisturbed) density of the medium.

Although Eqs. (2) and (3) have assumed a single bubble with a known location $\mathbf{r}$, the formulation can be readily extended to several bubbles of unknown locations by varying the putative source location $\mathbf{r}$ over a region of interest. In such a way, passive maps of source strength and source power can be obtained. With a linear array of aperture D placed on the x -axis and a normal distance z between the source and the array, the -3 dB transverse and axial resolutions of such maps are approximately¹²

$$\Delta x_{-3 \text{ dB}} = 0.89 f_{\#} \lambda, \quad (4)$$

$$\Delta z_{-3 \text{ dB}} = 6.95 f_{\#}^2 \lambda, \quad (5)$$

where $f_{\#} = z/D$, and λ is the mean wavelength of the source.

3. Methods

Cavitation was instigated in a 3% aqueous agar gel (degassed at 0.5 bar for 30 min prior to setting) using a spherically-focused HIFU transducer (H-102B SN22, Sonic Concepts, Bothell, WA) with a 1.06 MHz center frequency, 63 mm focal length, and 64 mm aperture. The HIFU transducer insonated the gel continuously at 1.06 MHz for 2 s. The -3 dB focal dimensions (12 mm axial length, 0.5 mm beam width) and the peak rarefactional focal pressure (PRFP) instigated were calibrated in water using a 0.5 mm needle hydrophone (SN1203, Precision Acoustics, Dorchester, UK). To avoid damaging the needle hydrophone, the maximum PRFP used in the calibrations was 1.1 MPa. Simulations of the non-linear propagation caused by the HIFU transducer have shown that extrapolating to higher pressures is valid up to 4 MPa PRFP,¹³ which is above the pressures used in this work. Three 2 s HIFU exposures were instigated in previously

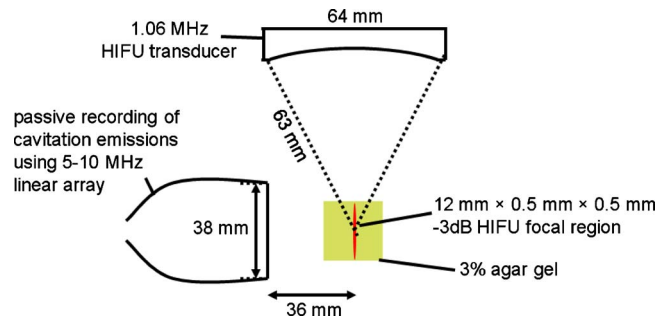


Fig. 1. (Color online) The experimental setup. A spherically focused transducer was used to generate continuous 2 s HIFU exposures in agar gel, at peak rarefactional focal pressures (PRFP) of 1.6 MPa, 2.4 MPa and 3.6 MPa. 64 elements from a linear array (chosen to span the entire aperture) were set to simultaneously record the emissions of cavitating bubbles in the focal region. The recordings were used to generate passive maps of source strength and source power using Eqs. (2) and (3), thereby providing a method of localizing cavitating bubbles and observing their behavior.

unexposed sections of the agar gel, which has a cavitation threshold of 1.2 MPa PRFP. The first exposure, at 1.6 MPa PRFP, was just above threshold, while the second and third exposures were well above threshold, at 2.4 MPa PRFP and 3.6 MPa PRFP respectively.

Throughout each HIFU exposure, a 38 mm aperture, 128-element, 5–10 MHz linear array (L10–5, Zonare Medical Systems, Mountain View, CA) was set to receive cavitation emissions passively, as depicted in Fig. 1. The linear array was connected to an ultrasound system (z.one with Research Package 2008 v3.5, Zonare Medical Systems, Mountain View, CA) capable of recording 64 channels of IQ data¹⁴ (7.14 MHz center frequency, 5 MHz bandwidth) simultaneously at a user-specified frame rate.¹⁵ In the current setup, the frame rate was set to 100 Hz, although judicious choice of the recording length of each frame would have allowed **frame rates of several kilohertz to be reached** over an exposure duration of several seconds. The IQ data were transferred to a USB memory device and converted to a Matlab (Mathworks, Natick, MA) data file for further processing. Each frame was upsampled to 200 MHz RF before applying Eqs. (2) and (3) (to the nearest time samples) to generate the passive maps.

In order to optimize image resolution, the recording channels were connected to elements spanning the entire aperture of the array.⁹ As shown in Fig. 1, the order of magnitude difference between the axial and transverse dimensions of the HIFU beam means that cavitation is expected to happen at a distance of 36 mm from the array (the location of the HIFU axis), with a greater portion of the HIFU axis exceeding the cavitation threshold as the incident pressure is increased. Therefore, with a 7.14 MHz center frequency and a speed of sound of 1482 m/s, Eqs. (4) and (5) predict axial and transverse resolutions of 0.2 mm, 1.3 mm. As a consequence, bubbles along the HIFU axis are expected to be resolved into 0.2 mm clusters of one or more bubbles.

Although this is the first publication to report the separation of cavitation emissions based on the location of origin, the reader is referred to prior work on passive cavitation mapping for further details of the experimental methods.⁹

4. Results

Figure 2 shows results obtained for the first, 1.6 MPa PRFP exposure, which was near the cavitation threshold of the agar gel. The first frame recorded after HIFU was switched on was used to generate a passive map of source power, which depicts three cavitating bubble clusters close to the HIFU focus. The temporal variation of source strength for the three clusters is also plotted. Since each HIFU cycle will reach bubbles closer to the HIFU transducer earlier, the temporal axis has been accordingly shifted so that emissions caused by the same HIFU wave-front can be identified. It is important to note that since attenuation and finite signal bandwidth

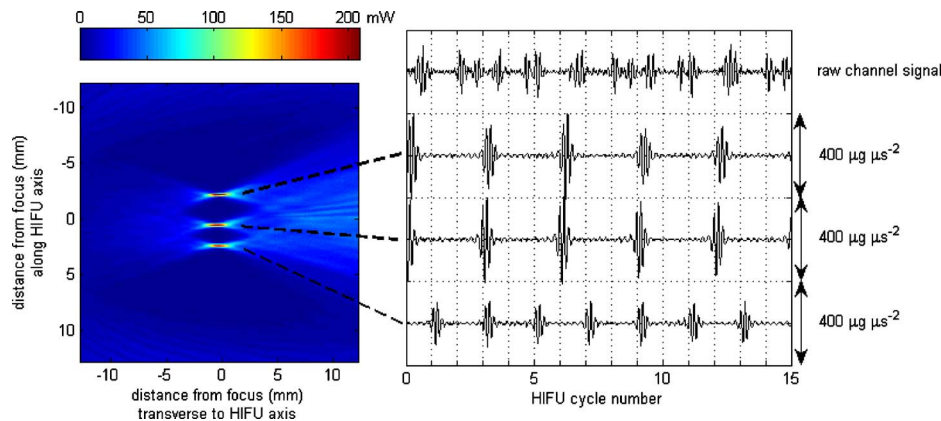


Fig. 2. (Color online) Passive source power map (left) and source strengths (right) generated from the first frame recorded after HIFU was turned on, at a setting of 1.6 MPa PRFP. The HIFU transducer is to the top of the source power map while the array is to its left, as depicted in Fig. 1. The source strength traces of the three bubbles show highly stable oscillations that are synchronous with each other. For comparison, the top trace on the right shows one of the raw channel signals from which the source strength traces were generated.

have not been accounted for, the source power and source strengths are underestimates of the true powers and source strengths of the bubble clusters, and finding the correct compensation factors has been left to future work.

The traces of source strength for the individual bubble clusters provide unique information about their activity. The two clusters closest to the HIFU transducer cavitate on every third HIFU cycle, while the distal cluster does this on every second cycle. These two clusters also cavitate in phase, that is, the cavitation is caused by the same HIFU cycle. Due to the respective period doubling and tripling, the distal cluster cavitates in phase with the other two every six cycles.

In the top-right corner of Fig. 2, one of the element signals used to generate the source traces is shown. It can be clearly seen that without further processing, the signal, showing a linear mixture of the source strengths above it, cannot be used to infer any of the information that has just been discussed. **This demonstrates the superiority of passive mapping over single element passive cavitation detection in assessing the stability and periodicity of cavitation.**

As the voltage fed to the HIFU transducer is increased, more and more bubbles will begin to cavitate along the HIFU axis. Since cavitation emissions may influence the cavitation of nearby bubbles, increased chaoticity of bubble behavior is thereby expected.¹⁶ This prediction is confirmed in Fig. 3, which shows distributions of source strengths along the HIFU axis

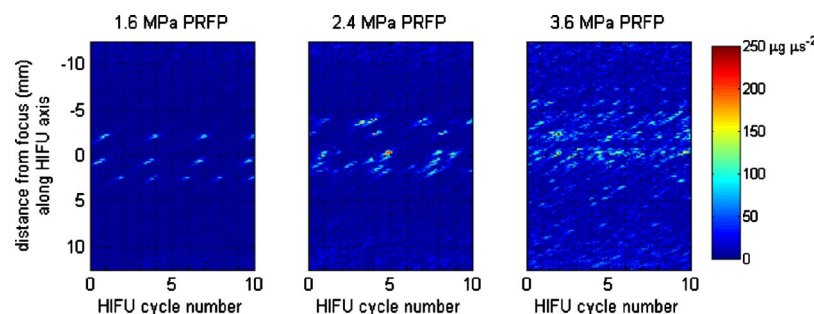


Fig. 3. (Color online) Source strength amplitudes along the HIFU axis for three exposure settings. Each image was generated from the first frame captured after HIFU was turned on. With an increasing region above the cavitation threshold, the number of cavitating bubbles increases and cavitation becomes more chaotic.

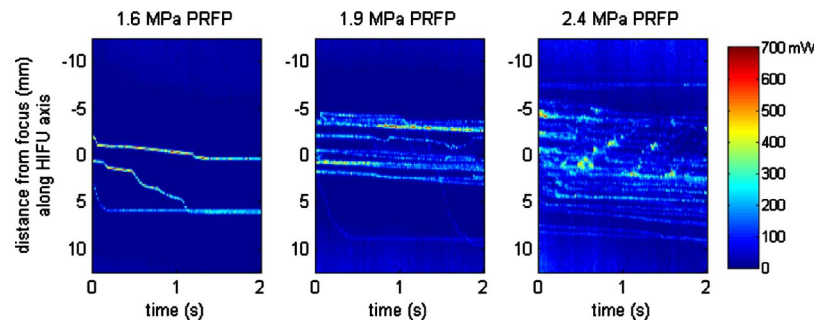


Fig. 4. (Color online) Passive source power maps along the HIFU axis during three 2 s HIFU exposures of increasing intensity. Bubbles tend to move away from the HIFU transducer, probably due to the acoustic radiation force and melting of the heated gel, and they are also seen to either coalesce or cluster with other bubbles.

for the three exposure settings.

Since the passive mapping system here presented can record frames over several seconds (at frame rates of 100 Hz and above), the system is able to monitor the evolution of cavitation emissions and locations over a timescale of seconds as well as microseconds. Figure 4 shows how passive maps generated along the HIFU axis vary during the three 2 s exposures, while Mm. 1 shows a movie of 2-D passive maps generated for the 1.6 MPa PRFP exposure. The maps all show a tendency for the bubbles to move away from the HIFU transducer, probably due to the acoustic radiation force generated by the HIFU insonation. The coalescence or clustering of bubbles is also observed.

Mm. 1. Passive cavitation maps generated from agar gel exposed to HIFU at 1.6 MPa PRFP for 2 s. Time from HIFU being on is indicated at the top. This is a file of type “avi” (3.7 Mbytes).

5. Discussion and conclusions

This work has shown how an array of receivers can be used to localize and separate emissions from simultaneously cavitating bubbles. Although the finite resolution of the imaging system meant that the identified source locations could have contained clusters of several bubbles, the periodic collapses shown on the emission traces makes it likely that the clusters behaved as single entities, a theory supported by other work.¹⁷

The technique presented in this paper enables, for the first time, localization and tracking of the evolution of multiple bubble clouds in optically opaque media, at a fraction of the cost required for high-speed photography of bubble dynamics in transparent media. The maps of source power and source strength provided, respectively, quantitative measures of the acoustic power radiated, and the rate of change of mass outflow (or source strength) generated by the bubble clusters. Future work will incorporate the effects of attenuation and recording bandwidth to provide corrected estimates of source strength, and will use a 2-D array to achieve 3-D mapping of bubble clouds.

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